Digital Level Regulator

1 Problem Description

The purpose is to implement a simple digital level regulator for an analog sinusoidal input signal. The system shall regulate the amplitude of the input signal from a control signal provided by a potentiometer and deliver a regulated analog output signal.

The overall problem is shown in Figure 1. The figure shows the level regulator as one system block with two inputs: the signal to be regulated and the potentiometer that determines the level.

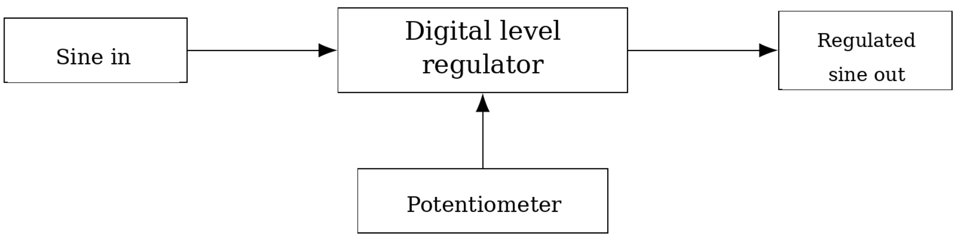

Figure 1: Overall problem figure for a digital level regulator.

2 Principle

2.1 Principle Schematic

The principle solution for the system in Figure 1 is to use two ADC channels and one DAC channel. One ADC channel reads the input signal, the other reads the control signal from the potentiometer, and the DAC delivers a new analog output signal.

The principle electrical wiring is shown in Figure 2. A signal generator delivers the sinusoidal input signal to one ADC input. The potentiometer is connected as a voltage divider between V_{DD} and ground, and the wiper is read on the other ADC input. The DAC output is measured with an oscilloscope.

Since the ADC input can only measure voltages within the reference range, a sine signal that is originally centered around 0 V must be shifted with a DC offset before it is sampled. In the signal model below, $x[n]$ is therefore treated as a digitized value within the ADC range.

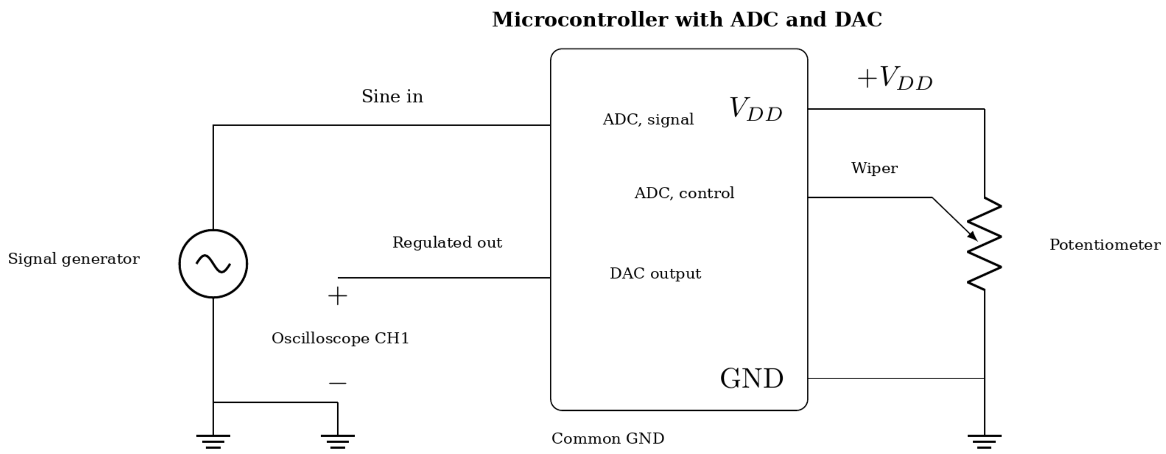

Figure 2: Principle circuit schematic for a digital level regulator with two ADC inputs and one DAC output.

2.2 Signal Model and Digital Level Regulation

If the digitized representation of the input signal is denoted $x[n]$, the output value is denoted $y[n]$, and the midpoint of the ADC and DAC range is denoted M, the following level regulation is used. Here, $g[n]$ is the level factor from the potentiometer, with $0 \leq g[n] \leq 1$.

$$y[n] = M + g[n]\big(x[n] - M\big). \tag{1}$$

Equation (1) means that the program does not scale the whole signal toward 0 V, but instead scales the deviation from the midpoint. This corresponds to regulating the AC signal swing around the midpoint after the signal has been moved into the valid voltage range of the ADC.

2.3 Potentiometer as Control Signal

The potentiometer is connected between V_{DD} and ground, and the wiper is read on a separate ADC channel. If the digitized control value is denoted $p[n]$, and the maximum digital value is denoted $P_{\max}$, the level factor can be written as

$$g[n] = \frac{p[n]}{P_{\max}}. \tag{2}$$

When Equation (2) is inserted into Equation (1), the result is

$$y[n] = M + \frac{p[n]}{P_{\max}}\big(x[n] - M\big). \tag{3}$$

This gives linear amplitude regulation without amplification above the input amplitude.

2.4 Program Structure

The program flow is based on reading one sample from the signal input and one sample from the potentiometer in each round. The deviation from the midpoint is then scaled, the result is limited to the valid DAC range, and the value is written to the DAC. This is a polling-based solution, meaning that the ADC values are read directly in the main loop without separate timing by interrupt. The following pseudocode describes the algorithm:

```
signal_value = read_ADC(signal_input)
potmeter     = read_ADC(potmeter_input)
midpoint     = M

deviation        = signal_value - midpoint
scaled_deviation = (deviation * potmeter) / P_max
output_value     = midpoint + scaled_deviation

output_value = limit(output_value, 0, P_max)

write_DAC(output, output_value)
```

3 Implementation and Test

3.1 Wiring

The practical implementation used Analog Discovery 2 (AD2) as signal source and oscilloscope, an AVR128DB48 Curiosity Nano board as the microcontroller platform, and a potentiometer as level control. The system supply was set to $V_{DD} = 3.3$ V.

Table 1 shows the concrete connections between the functions in Figure 2 and the physical implementation.

Table 1: Implemented wiring with AVR128DB48 Curiosity Nano, AD2, and potentiometer.

Function	Implemented connection
Signal input	AD2 Wavegen W1 to PD0/AIN0
Potentiometer	Wiper to PD1/AIN1, ends to $V_{DD} = 3.3$ V and GND
Analog output	PD6/DAC0 to AD2 Scope CH1
Reference	Common ground between AD2, Curiosity Nano, and potentiometer

The physical wiring is shown in Figure 3.

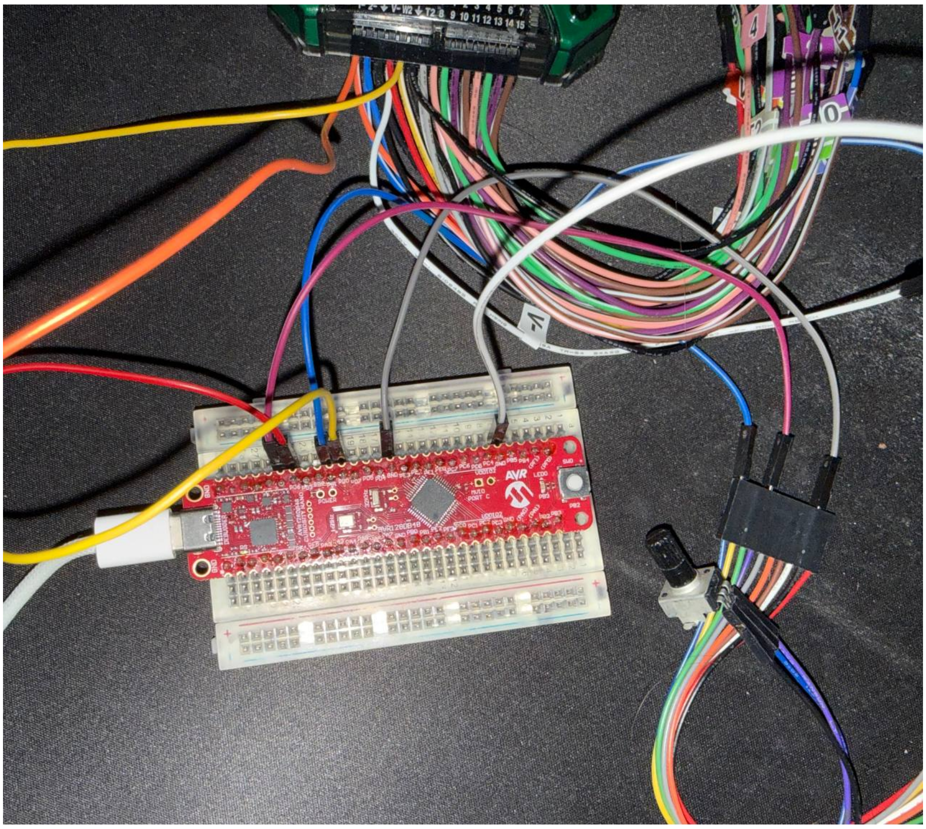

Figure 3: Physical wiring with AVR128DB48 Curiosity Nano, AD2 leads, and potentiometer.

3.2 Software

The software implements the algorithm from Section 2.4 with ADC and DAC driver functions. In the implementation, PD0/AIN0 was used as signal input, PD1/AIN1 as potentiometer input, and PD6/DAC0 as DAC output. Both ADC and DAC were used with 10-bit resolution, so the program uses $M = 512$ and $P_{\text{max}} = 1023$ in Equation (3).

3.3 Measurements

The measurements were performed with AD2 as both signal source and oscilloscope. Wavegen was used with sine shape, 100 Hz frequency (10 ms period), 0.5 V amplitude, approximately 1.0 Vpp at the input, 1.65 V DC offset, 50% symmetry, and phase 0°. This gave an input signal that varied between approximately 1.15 V and 2.15 V, inside the ADC range from 0 to 3.3 V. The DAC output on PD6 was observed with AD2 Scope, and the measurements were exported as CSV files for plotting.

Three operating points were documented: low, medium, and high potentiometer settings. The three measurement series are shown in Figures 4, 5, and 6.

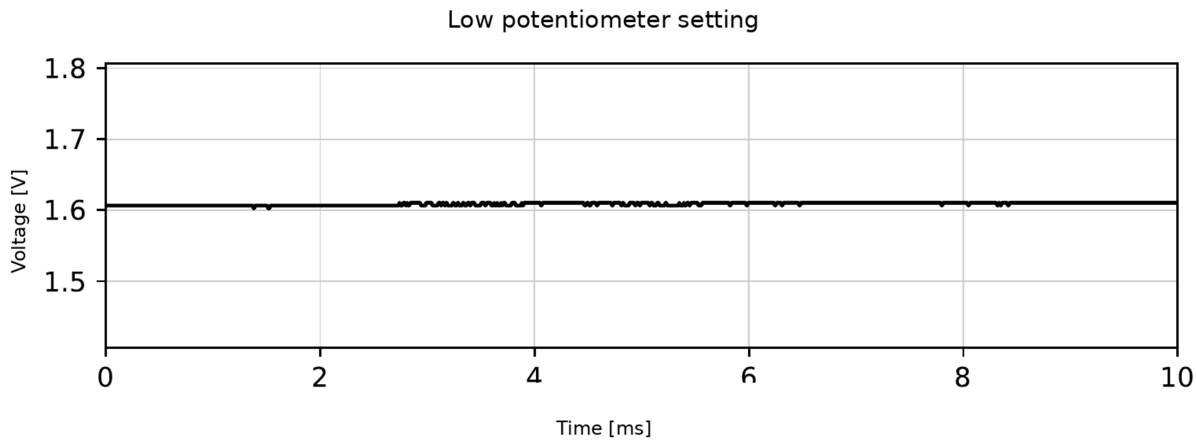

Figure 4: CSV-based plot for low potentiometer setting. The DAC output is almost flat around the midpoint.

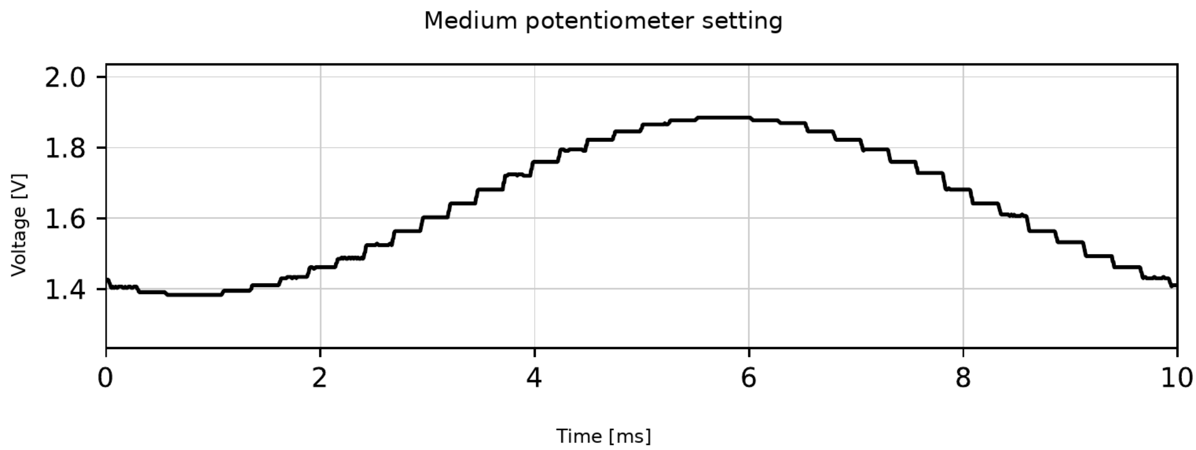

Figure 5: CSV-based plot for medium potentiometer setting. The DAC output follows the sine shape with reduced amplitude.

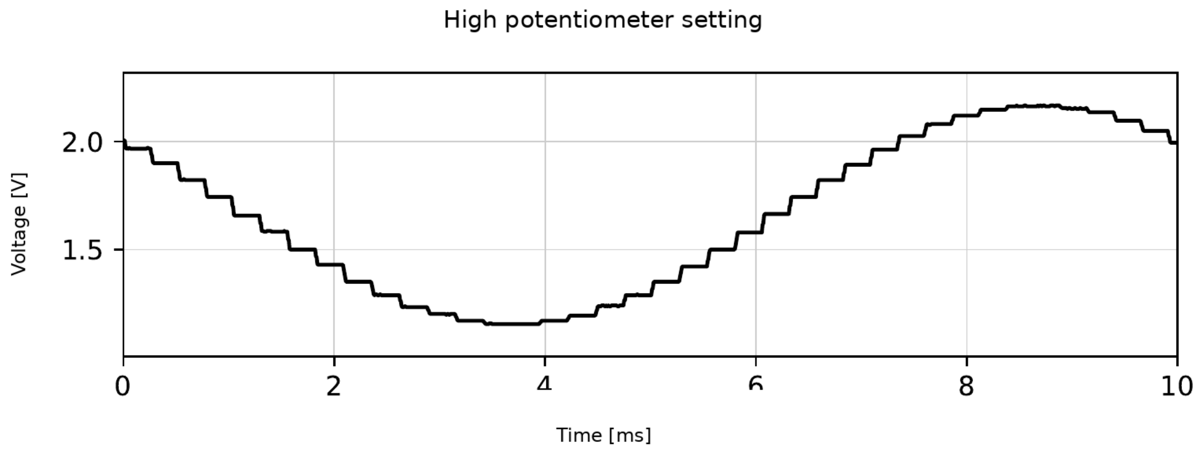

Figure 6: CSV-based plot for high potentiometer setting. The DAC output has a clearly larger amplitude.

The results show that the potentiometer regulates the amplitude of the DAC output. The approximate peak-to-peak values are 0.008 Vpp, 0.502 Vpp, and 1.011 Vpp for low, medium, and high potentiometer setting, respectively. The medium setting therefore gave approximately half the peak-to-peak value compared with the high setting, since $0.502/1.011 \approx 0.50$. At the low setting, the peak-to-peak value was almost zero, approximately $0.008/1.011 \approx 0.8\%$ of the maximum measured value. The observations show that the output in all three measurements stays around the same midpoint, while the amplitude changes. The DAC reconstruction also gives visible steps.

3.4 Evaluation of the Solution

In principle, the solution is feasible because the ADC digitizes the input signal, the program scales each sample, and the DAC writes a new analog level. The most important practical limitations are that the signal must have the correct DC offset before ADC reading, the ADC values must be read evenly enough relative to the signal frequency, and AD2, Curiosity Nano, and the potentiometer must share common ground. For the 100 Hz input signal, the sampling criterion requires a sampling frequency above 200 Hz. Based on the program code, each iteration contains two 10-bit ADC readings, two short delays during channel switching, the level calculation, DAC update, and a final delay of 50 μs. This gives a sampling frequency on the order of 13 kHz, which is well above the Nyquist limit and reduces the risk of aliasing for the selected test signal. If the input signal is not kept inside the valid ADC range, the signal can already get the wrong shape before level regulation is performed.

4 Conclusion

The project implemented a digital level regulator where the ADC reads both input signal and potentiometer value, the program scales the deviation from the midpoint, and the DAC delivers the regulated output signal. The measurements gave approximately 0.008 Vpp, 0.502 Vpp, and 1.011 Vpp at low, medium, and high potentiometer setting, respectively. This documents that the potentiometer regulates the signal amplitude from nearly zero to approximately the same peak-to-peak value as the input signal.

The limitation of the demonstration is the discrete reconstruction from the DAC, which gives visible steps on the output. For the selected test signal, the solution nevertheless satisfies the goal of digital level regulation.